



جامعة أم القرى
UMM AL-QURA UNIVERSITY

MOONLIT MARVELA

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Section Number: 2

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REVIEWS

The database is designed to manage hotel operations efficiently by organizing rooms, reservations, services, and billing. It ensures accurate tracking of room availability, reservation scheduling, and guest information while supporting additional services and flexible payment methods. This system enhances operational efficiency and improves the overall guest experience.

BUSINESS RULES:

Moonlit Marvela, a prestigious hotel, plans to develop a database system to efficiently manage information regarding its customers, staff, rooms, and services.

1. Reservation:

- Each reservation has a unique ID, check-in and check-out dates.
- Reservations are directly linked to specific customers and include one or more rooms.

2. Customer:

- Each customer is identified by a unique ID, along with their first name, last name, and phone number.
- Each email must be unique.

3. Room:

- Each room has a unique room ID, a room type (single, double, suite), its current status (available, reserved, or under maintenance), and price.
- A room can be reserved by multiple reservations at different times.

4. Bill:

- Each bill has a unique ID and includes room charges, service charges, and the payment method (cash or credit card).
- A bill is linked to one reservation through its reservation ID.
- A bill can include charges for one or more services requested by the customer during their stay.

5. Staff:

- Each staff member has a unique staff ID, name, position, phone number, email, working hours, and salary.
- Each staff member is assigned to a specific position and can only perform tasks related to their role (e.g., spa specialist, chef, driver).
- Each email must be unique.

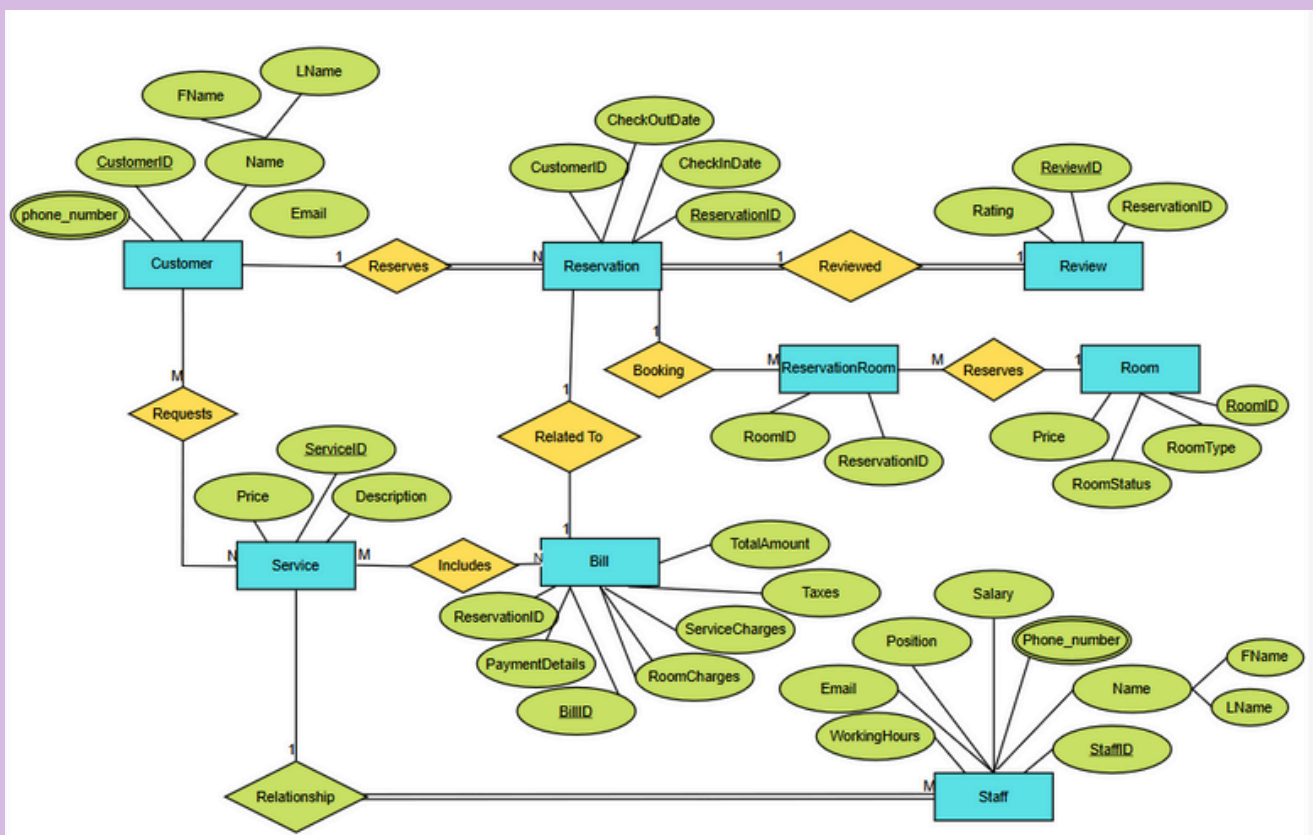
6. Service:

- Each service has a unique ID, description, and price.
- Services such as spa treatments, food delivery, and laundry are available to customers upon request.
- A service can be requested by one or more customers and is added to the customer's bill if used.

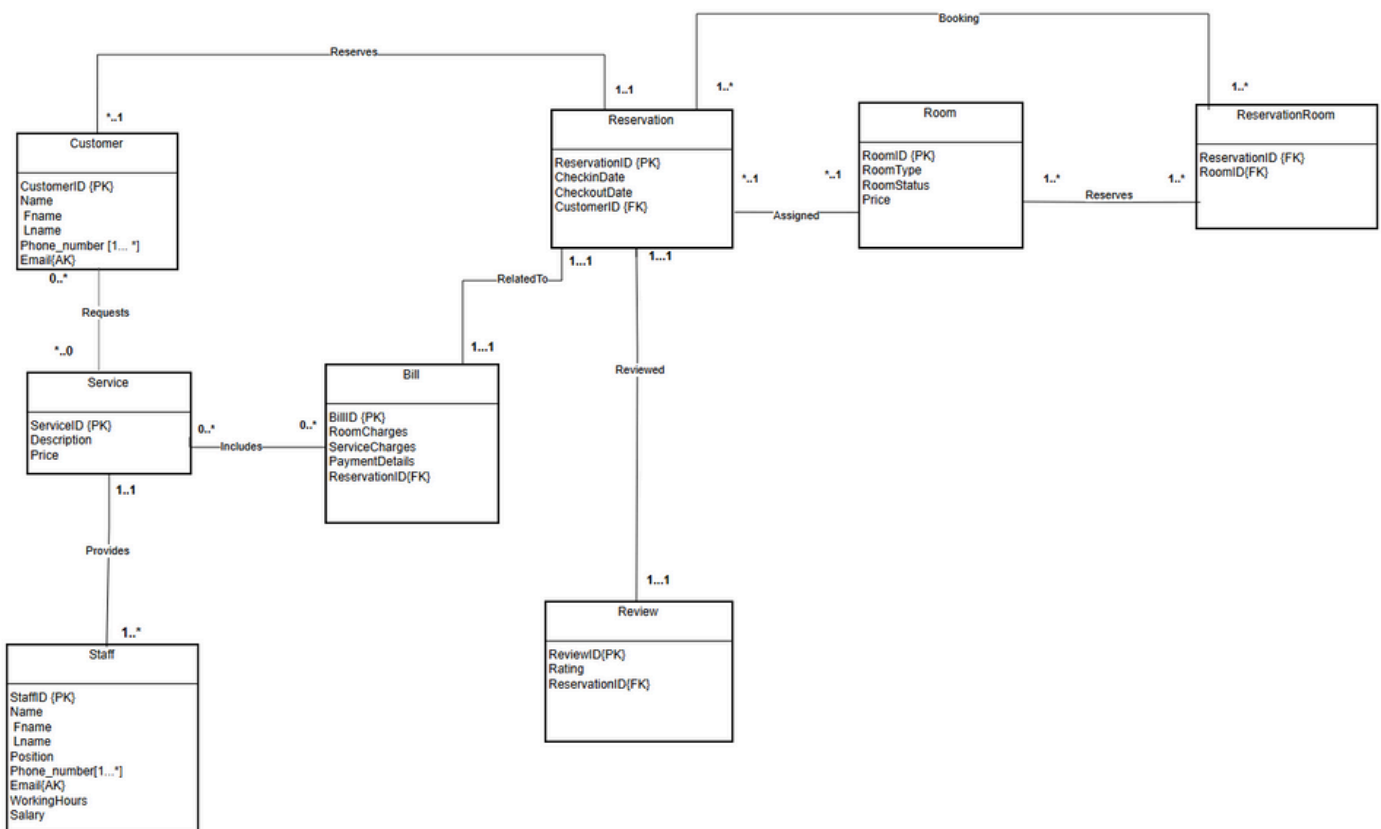
7. Review:

- Reviews have a unique review ID and a customer-provided rating.
- Every reservation must have exactly one review, and each review is tied to one specific reservation.

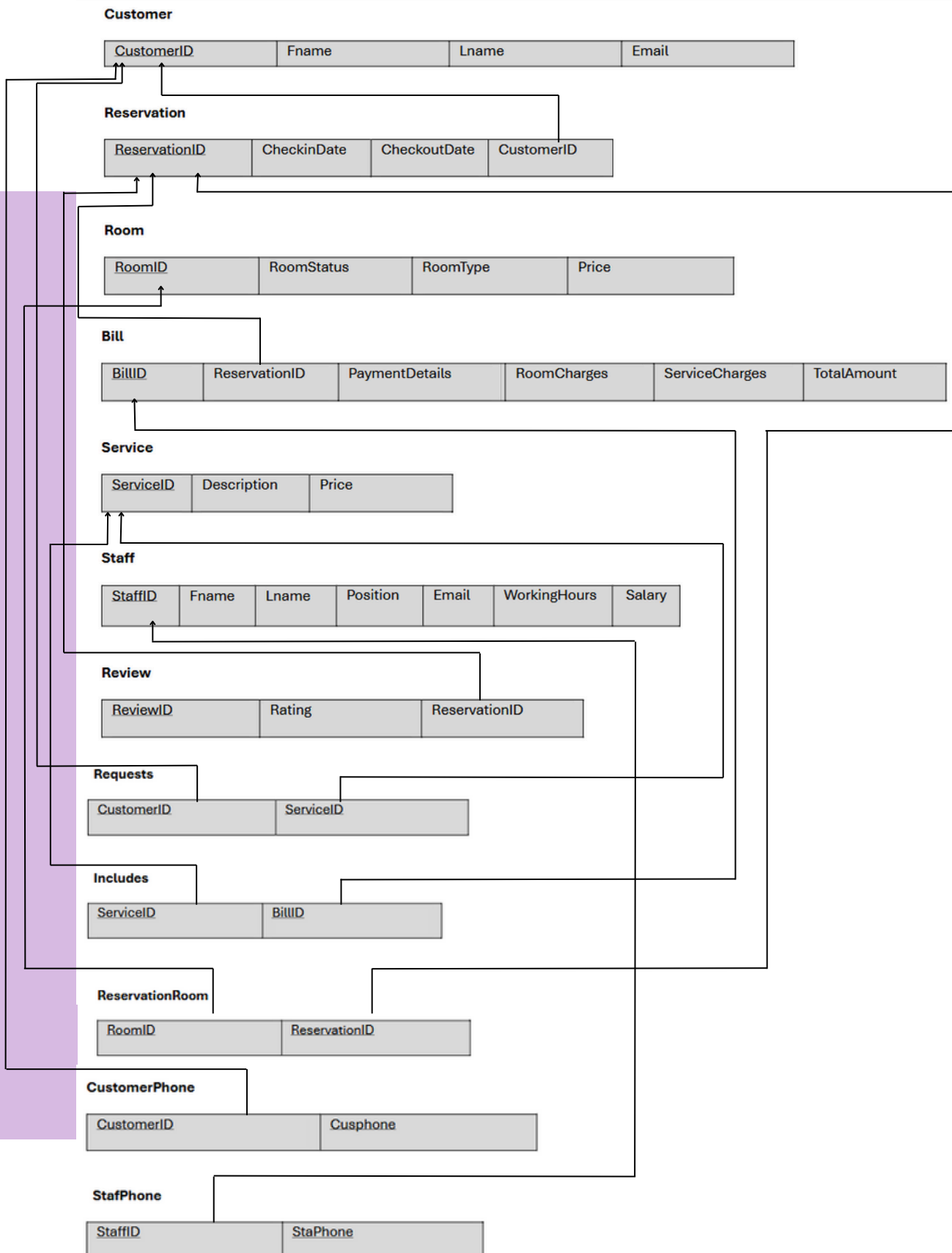
CHEN'S NOTATION :



UML DIAGRAM :



MAPPING OF ER INTO RELATIONAL SCHEMA:



TABLES :

```

IF EXISTS Customer (
    PRIMARY KEY,
    (0) NOT NULL,
    (0) NOT NULL,
    VARCHAR(15) NOT NULL,
    (0) UNIQUE NOT NULL

```

```

IF EXISTS Room (
    PRIMARY KEY CHECK (RoomID BETWEEN 1 AND 500),
    ('Single', 'Double', 'Suite') NOT NULL,
    ('Available', 'Reserved', 'Under Maintenance') DEFAULT 'Available',
    (0, 2) NOT NULL

```

```

CREATE TABLE IF NOT EXISTS Bill (
    BillID INT AUTO_INCREMENT PRIMARY KEY,
    ReservationID INT NOT NULL,
    RoomCharges DECIMAL(6, 2) NOT NULL,
    ServiceCharges DECIMAL(6, 2) DEFAULT 0.00,
    PaymentDetails VARCHAR(11) NOT NULL CHECK (PaymentDetails IN ('Cash', 'Credit Card')),
    FOREIGN KEY (ReservationID) REFERENCES Reservation(ReservationID)
);

```

```

CREATE TABLE IF NOT EXISTS Staff (
    StaffID INT PRIMARY KEY,
    fname VARCHAR(50) NOT NULL,
    lname VARCHAR(50) NOT NULL,
    Position VARCHAR(50) NOT NULL,
    Phone_number VARCHAR(15) NOT NULL,
    Email VARCHAR(50) UNIQUE NOT NULL,
    WorkingHours INT NOT NULL,
    Salary DECIMAL(7, 2) NOT NULL

```

```

CREATE TABLE IF NOT EXISTS Reservation (
    ReservationID INT AUTO_INCREMENT PRIMARY KEY,
    CheckinDate DATE NOT NULL,
    CheckoutDate DATE NOT NULL,
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)

```

```

CREATE TABLE IF NOT EXISTS ReservationRoom (
    ReservationID INT NOT NULL,
    RoomID INT NOT NULL,
    PRIMARY KEY (ReservationID, RoomID),
    FOREIGN KEY (ReservationID) REFERENCES Reservation(ReservationID),
    FOREIGN KEY (RoomID) REFERENCES Room(RoomID)

```

```

CREATE TABLE IF NOT EXISTS Services (
    ServiceID INT AUTO_INCREMENT PRIMARY KEY,
    Description VARCHAR(50) NOT NULL,
    Price DECIMAL(6, 2) NOT NULL
);

```

```

CREATE TABLE IF NOT EXISTS Includes (
    BillID INT NOT NULL,
    ServiceID INT NOT NULL,
    PRIMARY KEY (BillID, ServiceID),
    FOREIGN KEY (BillID) REFERENCES Bill(BillID),
    FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID)
);

```

```

CREATE TABLE IF NOT EXISTS Requests (
    CustomerID INT NOT NULL,
    ServiceID INT NOT NULL,
    PRIMARY KEY (CustomerID, ServiceID),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (ServiceID) REFERENCES Services(ServiceID)

```

INSERTION

```
INSERT INTO Room (RoomID, RoomType, RoomStatus, Price)
VALUES
```

```
(101, 'Single', 'Reserved', 350.00),
(102, 'Double', 'Reserved', 500.00),
(103, 'Suite', 'Reserved', 750.00),
(104, 'Single', 'Reserved', 350.00),
(105, 'Double', 'Reserved', 500.00),
(106, 'Suite', 'Reserved', 750.00),
(107, 'Single', 'Reserved', 350.00),
(108, 'Double', 'Reserved', 500.00),
(201, 'Single', 'Reserved', 350.00),
(202, 'Double', 'Reserved', 500.00),
(203, 'Suite', 'Available', 750.00),
(204, 'Suite', 'Available', 750.00),
(301, 'Single', 'Under Maintenance', 350.00),
(302, 'Double', 'Under Maintenance', 500.00),
(303, 'Suite', 'Under Maintenance', 750.00),
(304, 'Suite', 'Under Maintenance', 750.00);
```

```
INSERT INTO Bill (ReservationID, RoomCharges, ServiceCharges, PaymentDetails)
```

```
VALUES
```

```
(1, 850.00, 350.00, 'Credit Card'),
(2, 850.00, 150.00, 'Cash'),
(3, 350.00, 50.00, 'Credit Card'),
(4, 1250.00, 145.00, 'Cash'),
(5, 350.00, 200.00, 'Credit Card'),
(6, 500.00, 50.00, 'Cash'),
(7, 750.00, 0.00, 'Credit Card');
```

```
INSERT INTO Staff (StaffID, fname, lname, Position, Phone_number, Email, WorkingHours, Salary)
```

```
VALUES
```

```
(2000000001, 'Lama', 'Alghamdi', 'Receptionist', '0591234567', 'lama.alghamdi@gmail.com', 40, 4000.00),
(2000000002, 'Noura', 'Alotaibi', 'Housekeeping', '0592345678', 'noura.alotaibi@hotmail.com', 38, 1500.00),
(2000000003, 'Reem', 'Alzahrani', 'Gyn coach', '0593456789', 'reem.alzahrani@yahoo.com', 45, 5000.00),
(2000000004, 'Aseel', 'Alqahtani', 'Chef', '0594567890', 'aseel.alqahtani@outlook.com', 40, 8000.00),
(2000000005, 'Raghad', 'Alshahrani', 'Driver', '0595678901', 'raghad.alshahrani@gmail.com', 35, 3500.00),
(2000000006, 'Sarah', 'Alshehri', 'Spa Specialist', '0596789012', 'sarah.alshehri@hotmail.com', 30, 2500.00);
```

```
• INSERT INTO Reservation (CheckinDate, CheckoutDate, CustomerID)
```

```
VALUES
```

```
('2025-02-01', '2025-02-05', 1000000001),
('2025-03-05', '2025-03-10', 1000000001),
('2025-02-10', '2025-02-15', 1000000002),
('2025-03-01', '2025-03-05', 1000000003),
('2025-03-10', '2025-03-12', 1000000004),
('2025-04-01', '2025-04-03', 1000000005),
('2025-04-05', '2025-04-08', 1000000006);
```

```
• INSERT INTO ReservationRoom (ReservationID, RoomID)
```

```
VALUES
```

```
(1, 101),
(1, 102),
(2, 201),
(2, 202),
(3, 104),
(4, 105),
(4, 106),
(5, 107),
(6, 108),
(7, 103);
```

INSERTION

```
INSERT INTO Services (Description, Price)
VALUES
('Spa treatment', 200.00),
('Buffet', 150.00),
('Buffet for kids', 50.00),
('Laundry', 50.00),
('Gym', 75.00),
('Food Delivery', 70.00);
```

```
INSERT INTO Includes (BillID, ServiceID)
VALUES
(1, 1),
(1, 2),
(2, 3),
(4, 4),
(4, 5),
(5, 6);
```

- `INSERT INTO Requests (CustomerID, ServiceID)`
`VALUES`

```
(1000000001, 1),
(1000000001, 2),
(1000000002, 3),
(1000000003, 4),
(1000000003, 5),
(1000000004, 5),
(1000000005, 6);
```

- `INSERT INTO Review (ReservationID, Rating)`
`VALUES`

```
(1, 5),
(2, 4),
(3, 5),
(4, 3),
(5, 4),
(6, 5);
```


SELECT COMMANDS

```

17 • SELECT * FROM Room
18 WHERE RoomStatus = 'Available'
19 OR RoomStatus = 'Under Maintenance';
20

```

RoomID	RoomType	RoomStatus	Price
203	Suite	Available	750.00
204	Suite	Available	750.00
301	Single	Under Maintenance	350.00
302	Double	Under Maintenance	500.00
303	Suite	Under Maintenance	750.00
304	Suite	Under Maintenance	750.00
NULL	NULL	NULL	NULL

```

21 • SELECT
22     RoomStatus, COUNT(RoomID) AS TotalRooms
23 FROM Room
24 GROUP BY RoomStatus;
25

```

RoomStatus	TotalRooms
Reserved	10
Available	2
Under Maintenance	4

```

26 • SELECT
27     StaffID,
28     CONCAT (s.fname , ' ', s.lname) as Full_Name,
29     WorkingHours
30 FROM
31     Staff as s
32 HAVING
33     WorkingHours >= 35
34 ORDER BY WorkingHours DESC;
35

```

StaffID	Full_Name	WorkingHours
2000000003	Reem Alzahrani	45
2000000001	Lama Alghamdi	40
2000000004	Aseel Alqahtani	40
2000000002	Noura Alotaibi	38
2000000005	Raghad Alshahrani	35

SELECT COMMANDS

```

36
37 • SELECT -- Select (Order BY)
38     CONCAT (s.fname , ' ', s.lname) as Full_Name,
39     StaffID,
40     Salary,
41     Position
42 FROM Staff as s
43 ORDER BY Salary DESC;

```

	Full_Name	StaffID	Salary	Position
►	Aseel Alqahtani	2000000004	8000.00	Chef
	Reem Alzahrani	2000000003	5000.00	Gyn coach
	Lama Alghamdi	2000000001	4000.00	Receptionist
	Raghad Alshahrani	2000000005	3850.00	Driver
	Sarah Alshehri	2000000006	2500.00	Spa Specialist
	Noura Alotaibi	2000000002	1500.00	Housekeeping

```

IT (c.fname , ' ', c.lname) as Full_Name,
ne_number AS PhoneNumber

```

```
customer AS c
```

```
:
```

```
customerID IN (
```

```
SELECT
```

```
    r.CustomerID
```

```
FROM
```

```
    Reservation AS r
```

```
JOIN
```

```
    ReservationRoom AS rR ON r.ReservationID = rR .ReservationID
```

```
WHERE
```

```
    rR .RoomID = 101
```

```
);
```

	Full_Name	PhoneNumber
►	Rana Alnefaie	0501234567

SELECT COMMANDS

```

3
4 • SELECT
5     CONCAT(c.fname, ' ', c.lname) AS FullName,
6     c.Phone_number AS PhoneNumber,
7     r.ReservationID,
8     DATEDIFF(r.CheckoutDate, r.CheckinDate) AS DurationInDays,
9     SUM(ro.Price * DATEDIFF(r.CheckoutDate, r.CheckinDate)) AS FinalAmount
10  FROM
11     Customer AS c
12  JOIN
13     Reservation AS r ON c.CustomerID = r.CustomerID
14  JOIN
15     ReservationRoom AS rr ON r.ReservationID = rr.ReservationID
16  JOIN
17     Room AS ro ON rr.RoomID = ro.RoomID
18  GROUP BY
19     r.ReservationID, c.CustomerID, c.fname, c.lname, r.CheckinDate, r.CheckoutDate;
20

```

	FullName	PhoneNumber	ReservationID	DurationInDays	FinalAmount
▶	Rana Alnefaie	0501234567	1	4	3400.00
	Rana Alnefaie	0501234567	2	5	4250.00
	Sarah Almatrafi	0512345678	3	5	1750.00
	Tahani Albarqi	0523456789	4	4	5200.00
	Rawan Almitiri	0534567890	5	2	700.00
	Doha Alnabati	0545678901	6	2	1000.00
	Mihad Alahdal	0556789012	7	3	2250.00

```

32
33 • SELECT
34     BillID,
35     RoomCharges,
36     ServiceCharges,
37
38     (RoomCharges + ServiceCharges) AS TotalCharges,
39     (RoomCharges + ServiceCharges) * 0.15 AS Taxes,
40     (RoomCharges + ServiceCharges) + ((RoomCharges + ServiceCharges ) * 0.15 ) AS Price
41  FROM Bill;

```

	BillID	RoomCharges	ServiceCharges	TotalCharges	Taxes	Price
▶	1	850.00	350.00	1200.00	180.0000	1380.0000
	2	850.00	150.00	1000.00	150.0000	1150.0000
	3	350.00	50.00	400.00	60.0000	460.0000
	4	1250.00	145.00	1395.00	209.2500	1604.2500
	5	350.00	200.00	550.00	82.5000	632.5000
	6	500.00	50.00	550.00	82.5000	632.5000
	7	750.00	0.00	750.00	112.5000	862.5000

	RoomID	RoomType	RoomStatus	Price
	101	Single	Reserved	350.00
	102	Double	Reserved	500.00
	103	Suite	Reserved	750.00
	104	Single	Reserved	350.00
	105	Double	Reserved	550.00
	106	Suite	Reserved	750.00
	107	Single	Reserved	350.00
	108	Double	Reserved	500.00
	201	Single	Reserved	350.00
	202	Double	Reserved	500.00
	203	Suite	Available	750.00

StaffID	fname	lname	Position	Phone_number	Email	WorkingHours	Salary
2000000001	Lama	Alghamdi	Receptionist	0591234567	lama.alghamdi@gmail.com	40	4000.00
2000000002	Noura	Alotaibi	Housekeeping	0592345678	noura.alotaibi@hotmail.com	38	1500.00
2000000003	Reem	Alzahrani	Gyn coach	0593456789	reem.alzahrani@yahoo.com	45	5000.00
2000000004	Aseel	Alqahtani	Chief	0594567890	aseel.alqahtani@outlook.com	40	8000.00
2000000005	Raghad	Alshahrani	Driver	0595678901	raghad.alshahrani@gmail.com	35	3850.00
2000000006	Sarah	Alshehri	Spa Specialist	0596789012	sarah.alshehri@hotmail.com	30	2500.00

Tasks	student name
Mapping the ER into Relational Schema and reviewing the diagrams.	Duha
Coding and some commands.	Sara
Coding , Business roles, Prepared the report.	Rana
Chin's notation and help in UML.	Mehad
UML diagram , Prepared the report.	Tahani
Business roles and description of project	Rawan